

Fianlimab + cemiplimab in patients with advanced melanoma: Subgroup analyses by blinded independent central review

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Background

- Co-blockade of lymphocyte activation gene-3 (LAG-3)** may enhance the efficacy of **anti-programmed cell death-1 (PD-1)** therapies^{1,2}
- We have used VelocImmune technology³ to create fully human blocking antibodies to LAG-3⁴ and PD-1⁵
- Fianlimab** (anti-LAG-3) and **cemiplimab** (anti-PD-1) are high-affinity, fully human immunoglobulin G4 monoclonal antibodies^{3–5}
 - Fianlimab blocks LAG-3**/major histocompatibility complex class II-driven T-cell inhibition⁴
 - Cemiplimab blocks PD-1** binding to PD-ligand 1 and 2⁵ and is one of only two PD-1–blocking antibodies that is approved by the US Food and Drug Administration and European Medicines Agency for treatment of both squamous or non-squamous first-line advanced non-small cell lung cancer as either monotherapy or in combination with chemotherapy^{6,7}
- In the advanced melanoma setting, **fianlimab + cemiplimab combination demonstrated consistent and reproducible high clinical activity** (objective response rate [ORR]: 61%; median follow-up: 12.6 months; N=98) among anti-PD-1 naïve patients with a generally acceptable safety profile (NCT03005782)⁸

Objective

- Here, we present an efficacy analysis by blinded independent central review (BICR) with 23 months of median follow-up and safety data from subgroups of high-risk patients with advanced melanoma

Methods

An open-label, non-randomised, multi-cohort expansion study (NCT03005782)

Patients

Key inclusion criteria

- Metastatic or inoperable, locally advanced non-veal melanoma
- ≥18 years of age
- ECOG PS of 0 or 1
- ≥1 lesion measurable by RECIST v1.1

Key exclusion criteria

- Uveal melanoma
- Prior treatment with an anti-LAG-3 agent
- Prior treatment with an anti-PD-(L)1 agent for advanced melanoma
- Radiation therapy <2 weeks prior to enrolment

Endpoints

Primary endpoint

- ORR per RECIST v1.1

Secondary endpoints

- PFS
- DoR
- DCR
- Safety
- PK

Data cut-off date:
31 October 2023

Treatment

Fianlimab 1600 mg + cemiplimab 350 mg IV every 3 weeks, for up to 24 months

Initial cohort MM1 (n=40)*

Patients with advanced melanoma in the 1L or 2L setting who have never received anti-PD-(L)1

Confirmatory cohort MM2 (n=40)

Patients with advanced melanoma in the 1L setting who have never received anti-PD-(L)1

(Neo)adjuvant experienced cohort MM3 (n=18)*

Patients with advanced melanoma in the 1L setting who have received prior (neo)adjuvant systemic therapy,† including 13/18 patients who have received anti-PD-1

*23 NAP patients (Cohort MM1 [n=5] + Cohort MM3 [n=18]) received any (neo)adjuvant systemic therapy: anti-PD-1 (nivolumab or pembrolizumab); n=13; anti-BRAF/MEK: n=7; interferon: n=4; other: n=1; †Prior exposure to (neo)adjuvant systemic treatment (including anti-PD-1) with recurrence >6 months after adjuvant therapy. 1L, first-line; 2L, second-line; DCR, disease control rate; DoR, duration of response; ECOG PS, Eastern Cooperative Oncology Group performance status; IV, intravenous; LAG-3, lymphocyte activation gene-3; NAP, (neo)adjuvant pretreated; ORR, objective response rate; PD-(L)1, programmed cell death-(ligand)1; PFS, progression-free survival; PK, pharmacokinetics; RECIST, Response Evaluation Criteria in Solid Tumours.

AFFILIATIONS
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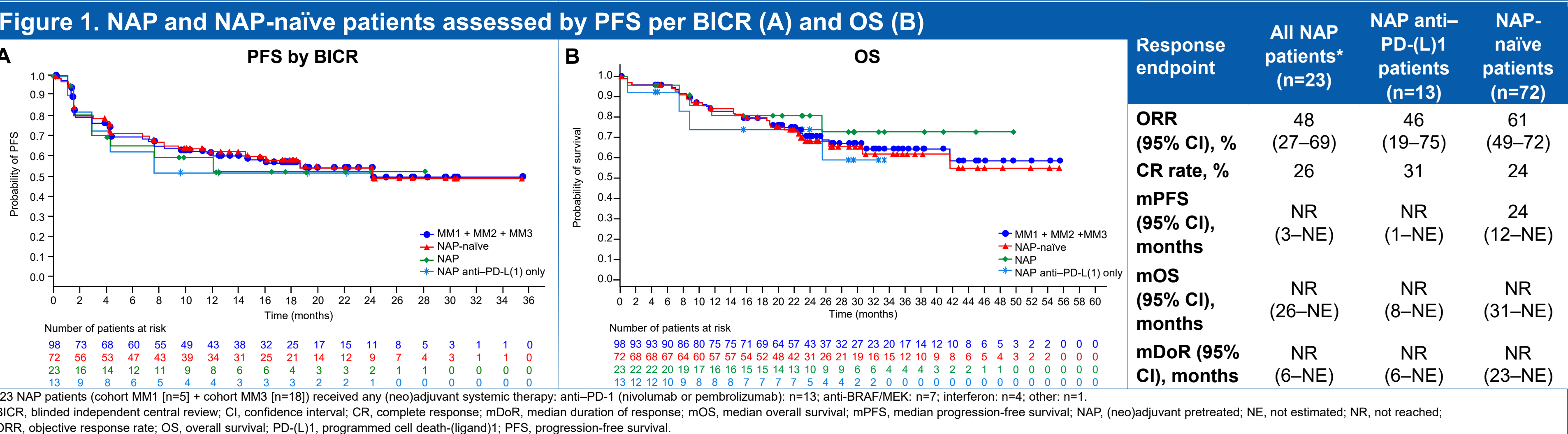
Results

- The median age across all cohorts (N=98) was 68 years (range 24–88); 60% of patients were men and 90% were White (**Supplementary Table 1**)
- Overall, complete response (CR) rate and ORR were 25% and 57% (95% confidence interval [CI] 47–67), respectively (**Table 1**)
- Median time to response and CR were 1.5 months and 4.1 months, respectively

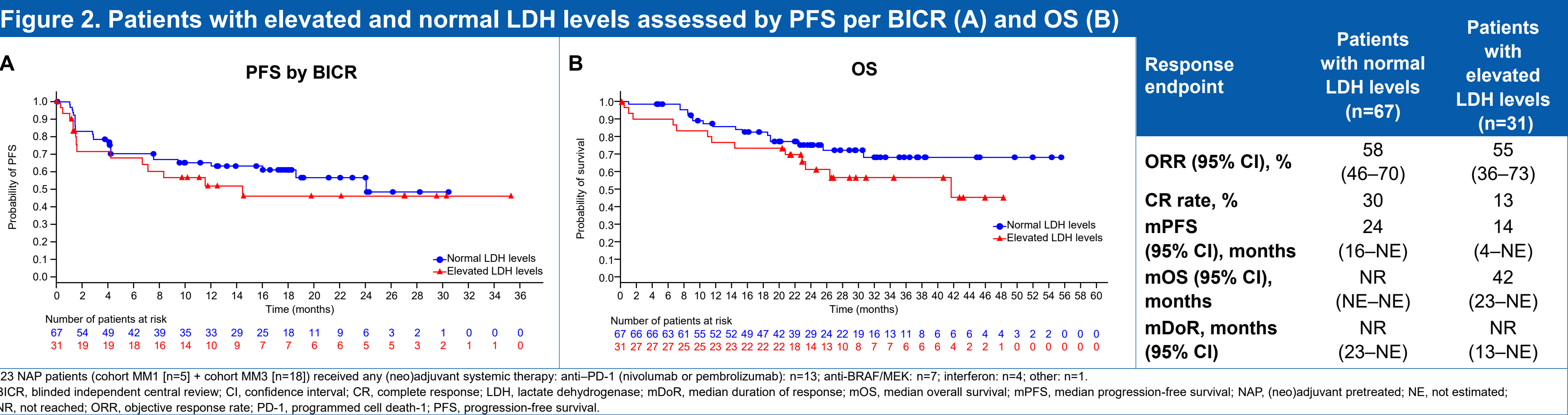
Response endpoint	Initial cohort MM1 (n=40)	Confirmatory cohort MM2 (n=40)	(Neo)adjuvant experienced cohort MM3 (n=18)	Combined cohorts MM1 + MM2 + MM3 (N=98)
Median follow-up (IQR), months	33 (13–42)	22 (17–24)	22 (9–29)	23 (14–31)
Median treatment exposure (IQR), weeks	37 (20–83)	36 (15–54)	28 (12–54)	36 (15–57)
KM-estimated mPFS (95% CI), months	NR (8–NE)	19 (8–NE)	12 (1–NE)	24 (12–NE)
12-month PFS (95% CI), %	62 (45–76)	64 (47–77)	42 (16–66)	60 (49–69)
24-month PFS (95% CI), %	55 (35–72)	47 (24–66)	42 (16–66)	49 (36–62)
mOS (95% CI), months	NR (42–NE)	NR (23–NE)	NR (26–NE)	NR (42–NE)
ORR (95% CI), %	NR (43–75)	63 (46–77)	39 (17–64)	57* (47–67)
mDoR (95% CI), months	NR (23–NE)	NR (14–NE)	NR (6–NE)	NR (23–NE)
DCR (95% CI), %	80 (64–91)	80 (64–91)	67 (41–87)	78 (68–85)
Best overall response, %				
CR	23	25	28	25
PR	38	38	11	33
SD	18	15	22	17
Non-CR/non-PD†	3	3	6	3
PD	15	15	17	15
Not evaluated‡	5	5	17	7

*The ORR in patients with acral/mucosal melanoma was 17% (1/6). †Non-CR/non-PD: only non-target lesions were present at baseline and did not reach CR or PD; ‡NE includes unevaluable, not applicable, unknown or missing tumour response. Six patients only had baseline tumour assessment, and one patient had no evidence of disease by BICR assessment. For individual patient data on NE and non-CR/non-PD, please refer to the supplementary information section. BICR, blinded independent central review; CI, confidence interval; CR, complete response; DCR, disease control rate; IQR, interquartile range; KM, Kaplan-Meier; mDoR, median duration of response; mOS, median overall survival; mPFS, median progression-free survival; NE, not estimated; NR, not reached; ORR, objective response rate; PD, progressive disease; PFS, progression-free survival; PR, partial response; SD, stable disease.

- ORR was 48% (95% CI 27–69) and 46% (95% CI 19–75) in all (neo)adjuvant pretreated (NAP) patients and NAP patients who had received prior anti-PD-1, respectively, compared with 61% (95% CI 49–72) in NAP-naïve patients (**Figure 1**)
- Despite differences in ORR between NAP and NAP-naïve patients, similar mPFS and median overall survival (mOS) were seen across these subpopulations



- Patients with a poor prognosis and lactate dehydrogenase (LDH) levels greater than the upper level of normal had similar ORR, mPFS and mOS compared to patients with normal LDH levels (**Figure 2**)



Conclusions

In this combined *post hoc* analysis of three independent cohorts:

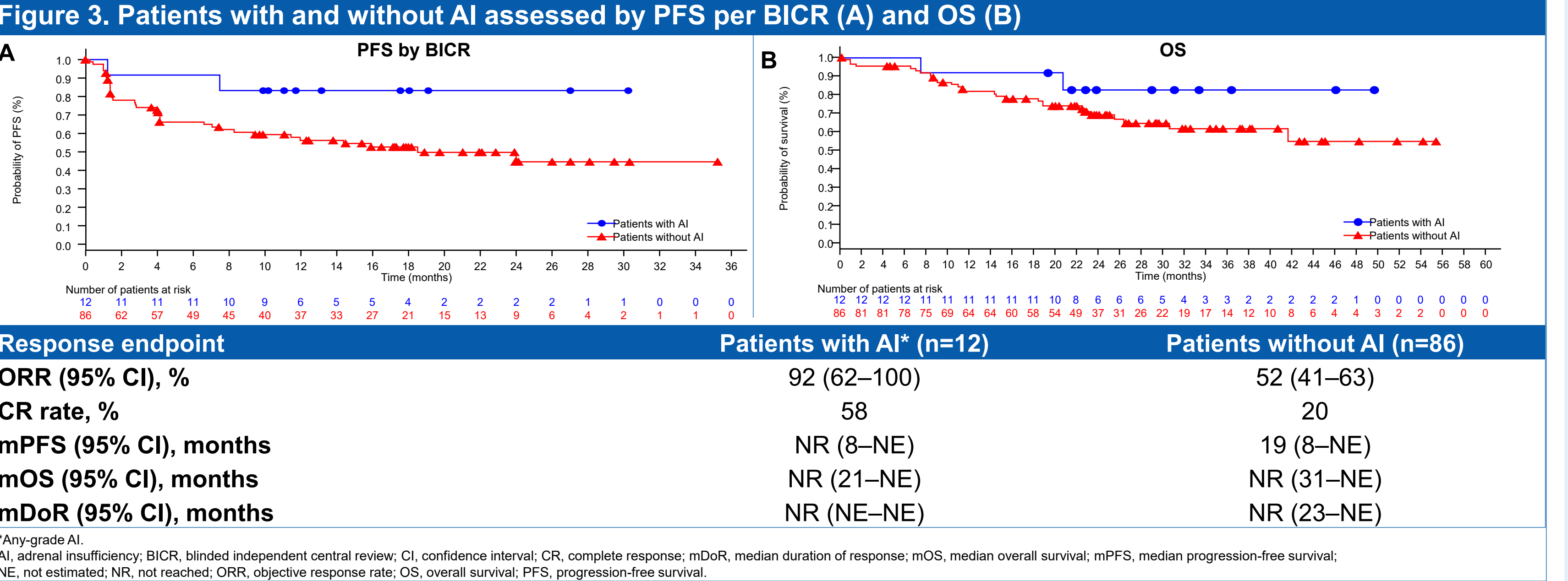
- This BICR analysis of fianlimab + cemiplimab combination with a median follow-up of 23 months continues to demonstrate high clinical activity and a generally acceptable safety profile in subgroups of patients with advanced melanoma
- Adrenal insufficiency (AI) is reported in 12% of patients treated with fianlimab + cemiplimab combination. Patients who experienced AI also experienced higher rates of overall and complete response.

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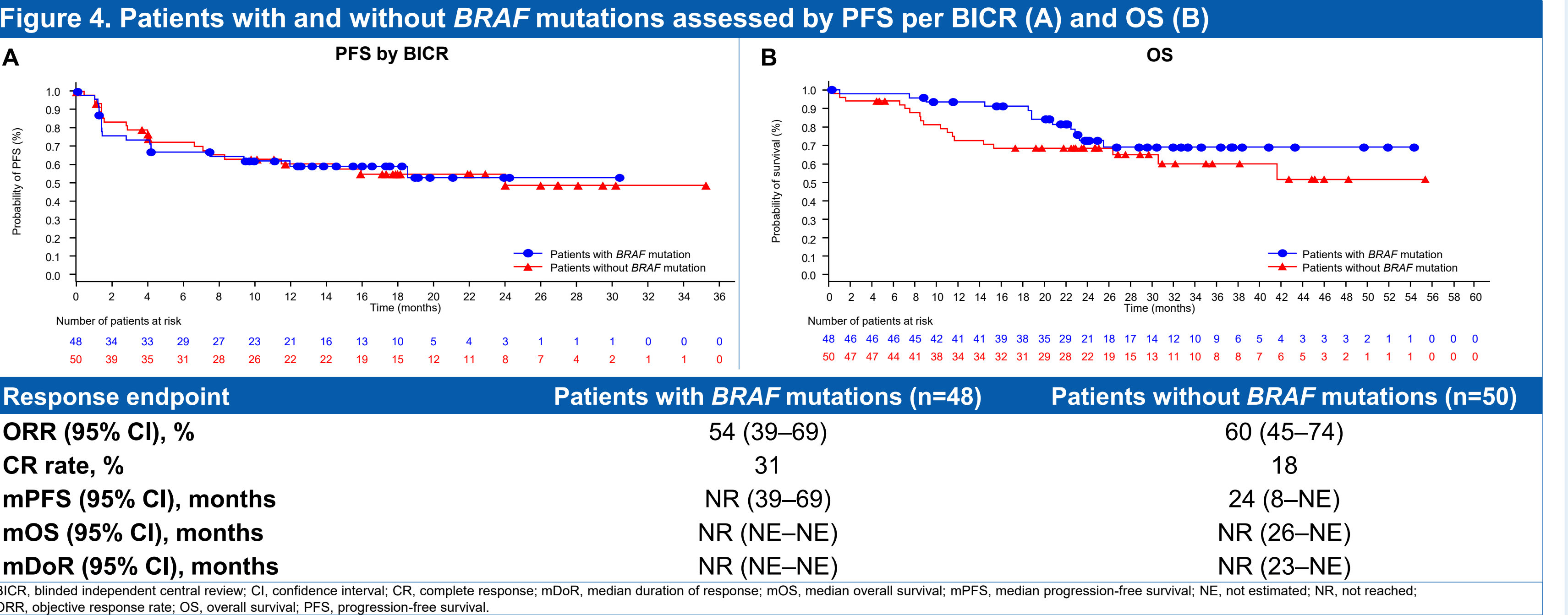


SUPPORTING MATERIALS
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- ORR was 92% (95% CI 62–100) and CR rate was 58% in 12 patients with any-grade treatment-related adrenal insufficiency (AI) (**Figure 3**). Patients who experienced treatment-related AI of any grade had higher ORR, PFS and OS benefit compared with patients who did not experience AI



- ORR was 54% (95% CI 39–69) in 48 patients with *BRAF* mutations; patients with *BRAF* mutations demonstrated a higher CR rate compared with patients without (**Figure 4**)
- mPFS and mOS were similar for all patients regardless of *BRAF* mutation status



Overall safety profile

- Across all three cohorts, the safety profile of fianlimab + cemiplimab combination was generally consistent with that of cemiplimab monotherapy and other anti-PD-(L)1 agents, except for higher rates of treatment-related AI (12% of patients; 5% with Grade 3 AI; **Table 2**)

Table 2. Combined cohort safety profile		
AEs	MM1 + MM2 + MM3 (N=98)	
	Any grade	Grade ≥3
TEAEs, regardless of attribution, %		
Overall	95	47*
Serious	39	36
TRAEs, %		
Overall	81	26†
Serious	21	19
Treatment-related immune-mediated AEs, %		
Treatment-related immune-mediated AEs in ≥10% of patients, %	39	13
AI‡	12	5§
Hypothyroidism	12	0

*TEAEs leading to death occurred in seven patients; †TRAEs leading to death occurred in two patients: one experienced colitis and one experienced cardiac shock. The patient who experienced cardiac shock also had COVID-19 with concurrent pulmonary oedema. ‡One additional patient experienced AI, which was assigned by the investigator as not related to drug treatment due to a pre-existing condition; §All five patients had Grade 3 AI. AE, adverse event; AI, adrenal insufficiency; TEAE, treatment-emergent adverse event; TRAE, treatment-related adverse event.

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Conclusions

In this combined *post hoc* analysis of three independent cohorts:

- With BICR analysis and a median follow-up of 23 months, fianlimab + cemiplimab combination continues to demonstrate high clinical activity and a generally acceptable safety profile in subgroups of patients with advanced melanoma
- Although adrenal insufficiency is reported in 12% of patients treated with fianlimab + cemiplimab combination, we report higher efficacy rates in patients who experienced AI than the overall study population

Supplementary Table 1. Baseline demographics and disease characteristics				
Characteristic	Initial cohort MM1* (n=40)	Confirmatory cohort MM2 (n=40)	(Neo)adjuvant experienced cohort MM3† (n=18)	Combined cohorts MM1 + MM2 + MM3 (N=98)
Median age (IQR), years	70 (55–77)	69 (58–76)	61 (50–70)	68 (55–75)
Male, %	63	58	61	60
White, %	90	90	89	90
Melanoma subtype, %				
Cutaneous non-acral	58	43	44	49
Acral	8	5	0	5
Mucosal	3	0	0	1
Missing	33	53	56	45
BRAF mutation, %	43	48	67	49
Metastatic status at baseline, %				
M0	13	5	22	11
M1‡	85	93	78	87
M1c	48	28	33	37
Unknown/missing	3	3	0	2
Median baseline tumour burden by BICR (IQR), mm§	56 (36–92)	45 (26–63)	48 (16–71)	48 (26–71)
LDH level >ULN, %	43	28	17	32
Liver metastases, %	35	10	11	20

*Five patients in cohort MM1 received prior adjuvant therapy; †17 patients in cohort MM3 received prior adjuvant therapy and one patient in cohort MM3 received prior (neo)adjuvant therapy; ‡M1a, b or c; §Sum of diameters of target lesion. 23 NAP patients (cohort MM1 [n=5] + cohort MM3 [n =18]) received any (neo)adjuvant systemic therapy: anti-PD-1 (nivolumab or pembrolizumab): n=13; anti-BRAF/MEK: n=7; interferon: n=4; other: n=1. BICR, blinded independent central review; IQR, interquartile range; LDH, lactate dehydrogenase; NAP, (neo)adjuvant pretreated; PD-1, programmed cell death-1; ULN, upper limit of normal.

Seven not evaluable patients

- One patient only had a baseline assessment, with no follow-up scan, and discontinued therapy, but was sent for BICR
- One patient received on-study radiotherapy and had a subsequent tumour assessment; the patient was excluded from the analysis due to the receipt of a new anticancer therapy
- One patient had no evidence of disease at all tumour assessments based on adjudication review
- Four patients only had screening images taken, with no follow-up scan, and discontinued therapy; therefore, they were not sent for BICR



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